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LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 30 September 2017

Untuk keadaan trafo bulan SEPTEMBER 2017

<u>1.0 Table of Contents</u>	<u>Page</u>
1.0 Specific Information	2
2.0 Our Comments and Recommendations of the Test Results Analysis	
- Transformer Unit MAIN TRAFO UNIT 1 (S/N: 100691) Test Certificate 4	3
- Transformer Unit TRAFO 1 ESP UNIT 1 (S/N: 10937) Test Certificate 5	4
- Transformer Unit TRAFO 2 ESP UNIT 1 (S/N: 10035) Test Certificate 6	5
- Transformer Unit TRAFO 3 ESP UNIT 1 (S/N: 10031) Test Certificate 7	6
- Transformer Unit TRAFO 4 ESP UNIT 1 (S/N: 10034) Test Certificate 8	7
- Transformer Unit MAIN TRAFO UNIT 3 (S/N: 1122170) Test Certificate 9	8
- Transformer Unit UAT 3 (S/N: 1035082) Test Certificate 10	9
- Transformer Unit TRAFO 1 ESP UNIT 3 (S/N: 0073084208) Test Certificate 11	10
- Transformer Unit TRAFO 2 ESP UNIT 3 (S/N: 0070084208) Test Certificate 12	11
- Transformer Unit TRAFO 3 ESP UNIT 3 (S/N: 0072084208) Test Certificate 13	12
- Transformer Unit TRAFO 4 UNIT 3 (S/N: 0071084208) Test Certificate 14	13
- Transformer Unit ASH HANDLING (S/N: 10820108) Test Certificate 15	14
- Transformer Unit ASH HANDLING (S/N: 10820107) Test Certificate 16	15
3.0 Summary of Recommendation	16

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	100691	4	Acidity, Dielectric, Color, Visualization and Water Content
2.	TRAFO 1 ESP UNIT 1	10937	5	Acidity, Dielectric, Color, Visualization and Water Content
3.	TRAFO 2 ESP UNIT 1	10035	6	Acidity, Dielectric, Color, Visualization and Water Content
4.	TRAFO 3 ESP UNIT 1	10031	7	Acidity, Dielectric, Color, Visualization and Water Content
5.	TRAFO 4 ESP UNIT 1	10034	8	Acidity, Dielectric, Color, Visualization and Water Content
6.	MAIN TRAFO UNIT 3	1122170	9	Acidity, Dielectric, Color, Visualization and Water Content
7.	UAT 3	1035082	10	Acidity, Dielectric, Color, Visualization and Water Content
8.	TRAFO 1 ESP UNIT 3	0073084208	11	Acidity, Dielectric, Color, Visualization and Water Content
9.	TRAFO 2 ESP UNIT 3	0070084208	12	Acidity, Dielectric, Color, Visualization and Water Content
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Acidity, Dielectric, Color, Visualization and Water Content
11.	TRAFO 4 UNIT 3	0071084208	14	Acidity, Dielectric, Color, Visualization and Water Content
12.	ASH HANDLING	10820108	15	Acidity, Dielectric, Color, Visualization and Water Content
13.	ASH HANDLING	10820107	16	Acidity, Dielectric, Color, Visualization and Water Content

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.014mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 82kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 8ppm is **ACCEPTABLE** while moisture in solid insulation of 0.57% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 1

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		45	NORMAL		GOOD	
31/07/2017	ONLINE	28	28				
30/08/2017	OFFLINE	49	49	NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	48		NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	41.5 AC	58 AC	0.5 AC	0.8691 AC	CLEAR AC	
31/07/2017	ROUTINE	0.017 AC		70 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		92 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		82 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 6 MONTHS:
THE DETECTION OF HYDROGEN IN CONJUNCTION WITH CARBON OXIDES MIGHT BE AN INDICATIVE OF LOW ENERGY DISCHARGE-CORONA INVOLVING SOLID INSULATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	10	0.82	A
31/07/2017	26	8	1.04	B
30/08/2017	34	6	0.49	A
29/09/2017	36	8	0.57	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 1

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.02	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.059mgKOH/g is **QUESTIONABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 58kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 19ppm is **ACCEPTABLE** while moisture in solid insulation of 1.80% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable Acidity and IFT values). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		36		NORMAL		GOOD	
31/07/2017	OFFLINE	28					
30/08/2017	OFFLINE	30				GOOD	NONE
29/09/2017	OFFLINE	35		NORMAL		GOOD	NONE

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.056 QU	25.3 UN	50 AC	1.5 AC	0.8529 AC	CLEAR AC	
31/07/2017	ROUTINE	0.059 QU		57 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.056 QU		65 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.059 QU		58 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	18	2.34	D
31/07/2017	25	20	2.60	D
30/08/2017	29	18	1.89	D
29/09/2017	32	19	1.80	D

FIELD SERVICE

DATE	SERVICE
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RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	95	ND	ND	ND	95

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.870 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.048mgKOH/g is **ACCEPTABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 59kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 19ppm is **ACCEPTABLE** while moisture in solid insulation of 1.80% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable IFT value). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		37		NORMAL		GOOD	
31/07/2017	OFFLINE						
30/08/2017	OFFLINE	39		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.042 AC	26.6 UN	41 AC	1.5 AC	0.8540 AC	CLEAR AC	
31/07/2017	ROUTINE	0.045 AC		25 QU	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		42 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.048 AC		59 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	27	21	2.48	D
31/07/2017	24	27	3.91	E
30/08/2017	28	14	1.65	C
29/09/2017	31	19	1.80	D

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.045mgKOH/gm is **ACCEPTABLE**. The levels remained stable without further oil degradation.
- **Dielectric** of 72kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 14ppm is **ACCEPTABLE** while moisture in solid insulation of 1.33% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		GOOD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	35		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	38		NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.048 AC	28.4 QU	44 AC	1.0 AC	0.8537 AC	CLEAR AC	
31/07/2017	ROUTINE	0.042 AC		77 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		61 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.045 AC		72 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	25	32	4.16	E
31/07/2017	25	16	2.08	D
30/08/2017	30	13	1.37	C
29/09/2017	32	14	1.33	B

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.031mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 53kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 15ppm is **ACCEPTABLE** while moisture in solid insulation of 1.23% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable IFT and Liquid Power Factor values). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		BAD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	34		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.028 AC	26.3 UN	24 UN	1.5 AC	0.8596 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		88 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		44 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.031 AC		53 AC	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	29	3.77	E
31/07/2017	26	15	1.95	D
30/08/2017	30	13	1.37	C
29/09/2017	33	15	1.23	B

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.05	0.02	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.08 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
6.	MAIN TRAFO UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.014mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 86kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 10ppm is **ACCEPTABLE** while moisture in solid insulation of 0.50% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46	56			GOOD	
31/07/2017	ONLINE	48	50				
30/08/2017	ONLINE	55	60	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE	58	68	NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	42.3 AC	78 AC	0.5 AC	0.8780 AC	CLEAR AC	
31/07/2017	ROUTINE	0.014 AC		77 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		86 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	24	10131	44084	11	641	2250	1	1	ND	678	57143

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
DGA SHOWS CARBON MONOXIDE LEVEL IS SLIGHTLY HIGH. OTHER GASES ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	15	1.23	B
31/07/2017	35	17	1.21	B
30/08/2017	35	8	0.57	A
29/09/2017	42	10	0.50	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.01	0.01	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.130 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.020mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 80kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 12ppm is **ACCEPTABLE** while moisture in solid insulation of 0.78% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		55			BAD	
31/07/2017	ONLINE						
30/08/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	60 AC	0.5 AC	0.8767 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		88 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		89 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.020 AC		80 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	20	16028	68308	20	971	3536	5	7	ND	1023	88895

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	11	0.90	A
31/07/2017	32	14	1.33	B
30/08/2017	35	11	0.78	A
29/09/2017	37	12	0.78	A

FIELD SERVICE
DATE SERVICE

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS.
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.02	0.01	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.390 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 45kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 20ppm is **QUESTIONABLE** while moisture in solid insulation of 1.64% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Based on the unfavorable IFT value from earlier oil analysis and current Water Content amount, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	23.2 UN	32 AC	1.5 AC	0.8432 AC	CLEAR AC	
31/07/2017	ROUTINE	0.028 AC		47 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		36 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.025 AC		45 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	36	26	1.85	D
31/07/2017	31	21	2.00	D
30/08/2017	30	22	2.31	D
29/09/2017	34	20	1.64	C

FIELD SERVICE
DATE SERVICE

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.115mgKOH/g is **UNACCEPTABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 21kV is **UNACCEPTABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour** of 2.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 37ppm is **UNACCEPTABLE** while moisture in solid insulation of 2.63% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** due to the unfavorable Acidity, IFT(from previous oil analysis), Dielectric, and Water Content. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.118 UN	16.2 UN	24 UN	3.0 AC	0.8435 AC	CLEAR AC	
31/07/2017	ROUTINE	0.112 UN		28 QU	3.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.109 UN		34 AC	2.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.115 UN		21 UN	2.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	39	2.54	D
31/07/2017	31	38	3.61	E
30/08/2017	26	17	2.21	D
29/09/2017	36	37	2.63	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - ZHEJIAN
DATE MFG - 2010
MAIN TANK S/N - **0070084208**
KVA -
PRI - 72000 V
SEC - 415 V
KILOGRAMS - 1350
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 731

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

ADDITIONAL EQUIPMENT

RADIATORS - YES
FANS - NO
H2O COOLED - NO
OIL PUMPS - NO
TOP FPV -
BOTTOM FPV -
BUCHOLZ - YES
CONSERVATOR - YES
LTC COMPARTMENT - NO
BUSHING LOC - TOP
FREE BREATHER - YES
HOSE LENGTH -
POWER AVAILABLE - NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 59kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 22ppm is **QUESTIONABLE** while moisture in solid insulation of 1.43% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Based on the unfavorable IFT value from earlier oil analysis and current Water Content amount, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE						
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.025 AC	27.8 UN	40 AC	1.5 AC	0.8415 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		66 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.025 AC		50 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.028 AC		59 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	27	1.75	D
31/07/2017	29	18	1.89	D
30/08/2017	35	21	1.49	C
29/09/2017	37	22	1.43	C

FIELD SERVICE
DATE SERVICE

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - ZHEJIAN
DATE MFG - 2010
MAIN TANK S/N - **0072084208**
KVA -
PRI - 72000 V
SEC - 415 V
KILOGRAMS - 1350
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 731

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

ADDITIONAL EQUIPMENT

RADIATORS - YES
FANS - NO
H2O COOLED - NO
OIL PUMPS - NO
TOP FPV -
BOTTOM FPV -
BUCHOLZ - YES
CONSERVATOR - YES
LTC COMPARTMENT - NO
BUSHING LOC - TOP
FREE BREATHER - YES
HOSE LENGTH -
POWER AVAILABLE - NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.022mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 59kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 15ppm is **ACCEPTABLE** while moisture in solid insulation of 1.58% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.020 AC	31.2 QU	62 AC	1.0 AC	0.8420 AC	CLEAR AC	
04/07/2017	ROUTINE	0.022 AC		60 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.020 AC		55 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		59 AC	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	32	19	1.80	D
04/07/2017	31	15	1.42	C
30/08/2017	27	17	2.01	D
29/09/2017	29	15	1.58	C

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	ND	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
12.	ASH HANDLING	10820108	15

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 57kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 16ppm is **ACCEPTABLE** while moisture in solid insulation of 1.31% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46					
31/07/2017	ONLINE						
30/08/2017	OFFLINE	30		NORMAL	FAIR		NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR		NONE

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	38 AC	1.0 AC	0.8779 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		56 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		53 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		57 AC	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	34	28	2.30	C
31/07/2017	26	21	2.73	D
30/08/2017	28	16	1.89	B
29/09/2017	34	16	1.31	B

FIELD SERVICE

DATE	SERVICE
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RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
13.	ASH HANDLING	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 28kV is **QUESTIONABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 31ppm is assessed as **QUESTIONABLE**.
Meanwhile, Water Content in solid insulation of 1.71% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content below 30ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Retest **Dielectric** and **Water Content** in 6 months to monitor their levels and to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

Page Disclaimer :

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DATE PRINTED 18/10/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		40				
31/07/2017	ONLINE						
30/08/2017	ONLINE	47		NORMAL	FAIR		NONE
29/09/2017	ONLINE	60		NORMAL	FAIR		NONE

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	37.3 AC	36 AC	1.0 AC	0.8788 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		40 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		31 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	29	21	2.20	C
31/07/2017	31	32	3.04	D
30/08/2017	34	29	2.38	C
29/09/2017	40	31	1.71	B

FIELD SERVICE

DATE	SERVICE
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(Formerly known as Myers-Musteq Sdn Bhd)

DATE PRINTED 18/10/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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3.0 Summary of Our Comments & Recommendations:

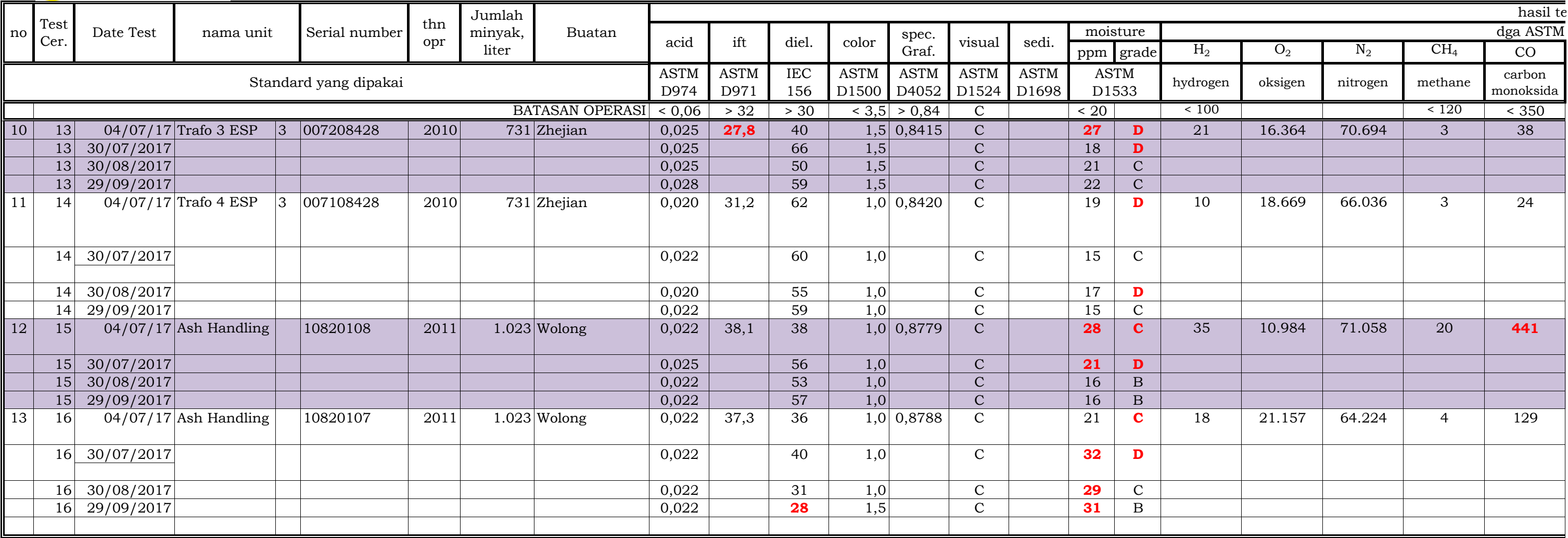
No.	TX Unit No.	Serial No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	100691	4	• Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	• Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	• Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	• Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	• Perform OIL REGENERATION
6.	MAIN TRAFO UNIT 3	1122170	9	• Periodic condition monitoring
7.	UAT 3	1035082	10	• Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	• Perform OIL REGENERATION
9.	TRAFO 2 ESP UNIT 3	0070084208	12	• Perform OIL REGENERATION
10.	TRAFO 3 ESP UNIT 3	0072084208	13	• Perform OIL REGENERATION
11.	TRAFO 4 UNIT 3	0071084208	14	• Periodic condition monitoring
12.	ASH HANDLING	10820108	15	• Periodic condition monitoring
13.	ASH HANDLING	10820107	16	• Retest 6 months: Dielectric and Water Content • Other parameters: Periodic condition monitoring

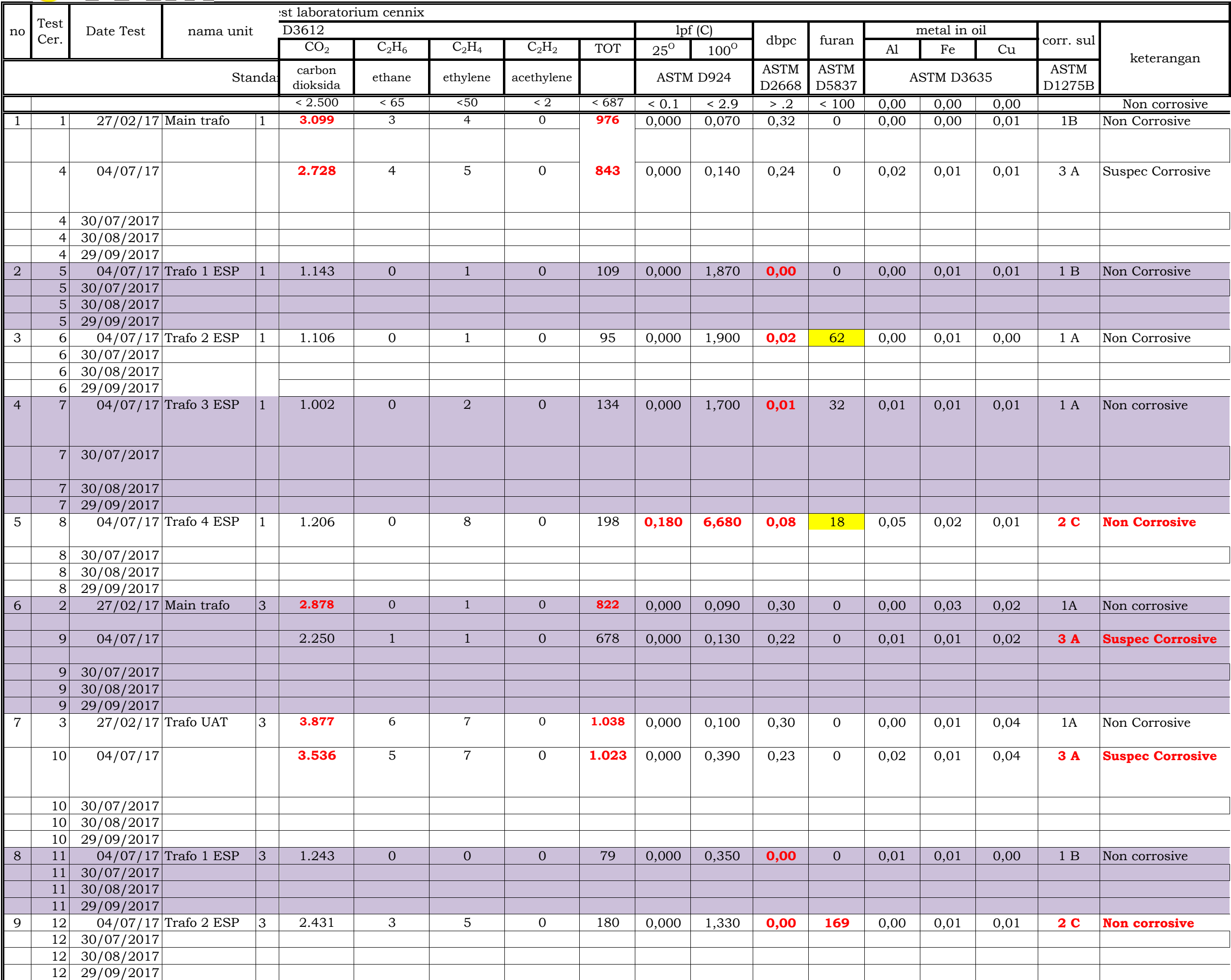
Page Disclaimer :

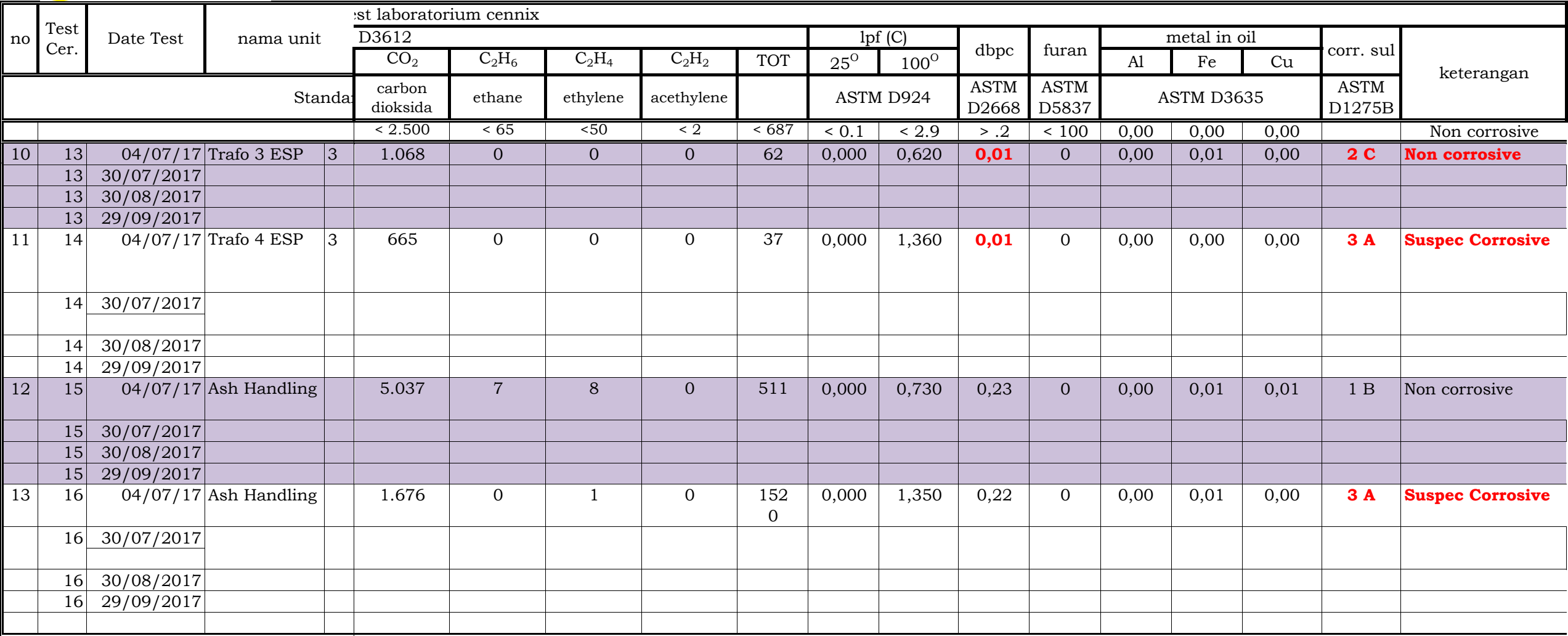
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no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil tes														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
1	1	27/02/17	Main trafo	1	1006901	2011	30.857	Wolong	0,014	39,9	99	0,5	0,8693	C		9	A	135	15.834	57.625	27	807
	4	04/07/17							0,017	41,5	58	0,5	0,8691	C		10	A	147	9.471	4.328	32	655
	4	30/07/2017							0,017		70	0,5		C		8	B					
	4	30/08/2017							0,017		92	0,5		C		6	A					
	4	29/09/2017							0,014		82	0,5		C		8	A					
2	5	04/07/17	Trafo 1 ESP	1	10937	2010	1.143	Fujia	0,056	25,3	50	1,5	0,8529	C		18	D	35	20.494	67.487	4	69
	5	30/07/2017							0,059		57	1,5		C		20	D					
	5	30/08/2017							0,056		65	1,5		C		18	D					
	5	29/09/2017							0,059		58	1,5		C		19	D					
3	6	04/07/17	Trafo 2 ESP	1	10035	2010	1.143	Fujia	0,042	26,6	41	1,5	0,8540	C		21	D	27	18.639	66.781	3	64
	6	30/07/2017							0,045		25	1,5		C		27	E					
	6	30/08/2017							0,042		42	1,5		C		14	C					
	6	29/09/2017							0,048		59	1,5		C		19	D					
4	7	04/07/17	Trafo 3 ESP	1	10031	2010	1.143	Fujia	0,048	28,4	44	1,0	0,8537	C		32	E	32	15.273	65.111	5	95
	7	30/07/2017							0,042		77	1,5				10	D					
	7	30/08/2017							0,042		61	1,5		C		13	C					
	7	29/09/2017							0,045		72	1,5		C		14	B					
5	8	04/07/17	Trafo 4 ESP	1	10034	2010	1.143	Fujia	0,028	26,3	24	1,5	0,8596	C		29	E	13	12.265	66.770	5	172
	8	30/07/2017							0,022		88	1,5		C		15	D					
	8	30/08/2017							0,028		44	1,5		C		13	C					
	8	29/09/2017							0,031		53	1,0		C		15	B					
6	2	27/02/17	Main trafo	3	1122170	2010	26.706	Shandong	0,017	39,7	87	0,5	0,8781	C		9	A	4	13.330	43.140	9	808
	9	04/07/17							0,017	42,3	78	0,5	0,8780	C		15	B	24	10.131	44.084	11	641
	9	30/07/2017							0,014		77	0,5		C		17	B					
	9	30/08/2017							0,017		86	0,5		C		8	A					
	9	29/09/2017							0,014		86	0,5		C		10	A					
7	3	27/02/17	Trafo UAT	3	1035082	2010	4.749	Shandong	0,017	37,6	94	0,5	0,8768	C		9	A	20	12.535	56.879	20	985
	10	04/07/17							0,022	38,1	60	0,5	0,8767	C		11	A	20	16.028	68.308	20	971
	10	30/07/2017							0,025		88	0,5		C		14	B					
	10	30/08/2017							0,022		89	0,5		C		11	A					
	10	29/09/2017							0,020		80	0,5		C		12	A					
8	11	04/07/17	Trafo 1 ESP	3	007308428	2010	731	Zhejiang	0,022	23,2	32	1,5	0,8432	C		26	D	14	12.943	68.807	3	62
	11	30/07/2017							0,028		47	1,5		C		21	D					
	11	30/08/2017							0,028		36	1,5		C		22	D					
	11	29/09/2017							0,025		45	1,5		C		20	C					
9	12	04/07/17	Trafo 2 ESP	3	007008428	2010	731	Zhejiang	0,118	16,2	24	3,0	0,8435	C		39	D	38	16.781	70.523	5	129
	12	30/07/2017							0,112		28	3,0		C		38	E					
	12	30/08/2017							0,109		34	2,5		C		17	D					
	12	29/09/2017							0,115		21	2,5		C		37	D					









no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
1	1	27/02/17	Main trafo	1	Retest 6 months: DGA Check leakages Other parameters: Periodic Condition Monitoring	FOR TEST 29 SEPTEMBER 2017 Yang perlu diganti minyak hanyalah Tc 12. Karena minyak sedikit, jauh lebih murah dibanding regenerasi. Yang perlu dipurifikasi adalah Tc 5, 6,11, 14, 16. Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13,14) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami sat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg pelaksanaan diatas 7 cycle. Dari hasil terakhir nampak perubahan moisture pada isoilasi ("Grade") pada trafo2 ESP naik turun ini menandakan isolasi sudah mulai kurang bagus. FOR TEST 30 AUGUST 2017 Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13,14) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami sat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg pelaksanaan diatas 7 cycle Dari hasil terakhir nampak perubahan moisture pada isoilasi ("Grade") pada trafo2 ESP naik turun ini menandakan isolasi kurang bagus. FOR TEST 30 JULY 2017 Untuk trafo ESP 2 Unit 3 sebaiknya minyaknya segera diganti saja, karena Acidity dan Grade Water Content sudah sangat di luar standard Untuk trafo Ash Handling , segera dipurifikasi karena water content sudah jauh di atas standard. FOR TEST 04 JULY 2017 Untuk Main Trafo 1 dan 3 serta Trafo UAT 3 yang tinggi CO dan CO ₂ nya, harap diperhatikan masalah cooling system antara lain menyalakan fan sejak awal trafo dinyalakan. Salah satu akibat dari CO dan CO ₂ yang tinggi (yang artinya sedang mengalami gangguan thermal secara internal) adalah penurunan kondisi Corrosive Sulphur. Dari data yang ada, pada bulan February yang lalu, masih "Non Corrosive", dan pada bulan Juli ini menjadi "Suspec Corrosive". Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13) yang kurang
	4	04/07/17			Retest 6 months: DGA Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	4	30/07/2017			Periodic condition monitoring	
	4	30/08/2017			Periodic condition monitoring	
	4	29/09/2017			Periodic condition monitoring	
2	5	04/07/17	Trafo 1 ESP	1	Perform OIL REGENERATION	
	5	30/07/2017			Perform OIL REGENERATION	
	5	30/08/2017			Perform OIL REGENERATION	
	5	29/09/2017			Perform OIL REGENERATION	
3	6	04/07/17	Trafo 2 ESP	1	Perform OIL REGENERATION	
	6	30/07/2017			Perform OIL REGENERATION	
	6	30/08/2017			Perform OIL REGENERATION	
	6	29/09/2017			Perform OIL REGENERATION	
4	7	04/07/17	Trafo 3 ESP	1	Retest 6 months: Acidity, IFT, LPF, Diel and Water Content Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	7	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	7	30/08/2017			Periodic condition monitoring	
	7	29/09/2017			Periodic condition monitoring	
5	8	04/07/17	Trafo 4 ESP	1	Perform OIL REGENERATION Inspect silica gel	
	8	30/07/2017			Perform OIL REGENERATION	
	8	30/08/2017			Perform OIL REGENERATION	
	8	29/09/2017			Perform OIL REGENERATION	
6	2	27/02/17	Main trafo	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	
	9	04/07/17			Add Passivator	
					Periodic Condition Monitoring	
	9	30/07/2017			Periodic condition monitoring	
	9	30/08/2017			Periodic condition monitoring	
	9	29/09/2017			Periodic condition monitoring	
7	3	27/02/17	Trafo UAT	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	
	10	04/07/17			Add Passivator Periodic Condition Monitoring Inspect silica gel	
	10	30/07/2017			Periodic condition monitoring	
	10	30/08/2017			Periodic condition monitoring	
	10	29/09/2017			Periodic condition monitoring	
8	11	04/07/17	Trafo 1 ESP	3	Perform OIL REGENERATION	
	11	30/07/2017			Perform OIL REGENERATION	
	11	30/08/2017			Perform OIL REGENERATION	
	11	29/09/2017			Perform OIL REGENERATION	
9	12	04/07/17	Trafo 2 ESP	3	Perform OIL REGENERATION	
	12	30/07/2017			Perform OIL REGENERATION	
	12	30/08/2017			Perform OIL REGENERATION	
	12	29/09/2017			Perform OIL REGENERATION	



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
10	13	04/07/17	Trafo 3 ESP	3	Perform OIL REGENERATION	Mengingat banyak trafo ESP (Tc 6,7,8,10,11,12,16) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 65 ⁰ C dan vacuum diatas 74 CMHg diatas 7 cycle Trafo ESP (Tc 6,7,8) Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus Untuk trafo Ash Handling sebaiknya segera dipurifikasi juga dengan temperatur diatas 65 ⁰ c dan vacuum diatas 74 CMHg sekitar >7 cycle
	13	30/07/2017			Perform OIL REGENERATION	
	13	30/08/2017			Perform OIL REGENERATION	
	13	29/09/2017			Perform OIL REGENERATION	
11	14	04/07/17	Trafo 4 ESP	3	Retest 6 months: Acidity, IFT and LPF Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	14	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	14	30/08/2017			Periodic condition monitoring	
	14	29/09/2017			Periodic condition monitoring	
12	15	04/07/17	Ash Handling		Add Passivator Periodic Condition Monitoring	
	15	30/07/2017			Periodic condition monitoring	
	15	30/08/2017			Periodic condition monitoring	
	15	29/09/2017			Periodic condition monitoring	
13	16	04/07/17	Ash Handling		Add Passivator Periodic Condition Monitoring	
	16	30/07/2017			Retest 6 months: Dielectric and Water Content Other parameters: Periodic Condition Monitoring	
	16	30/08/2017			Periodic Condition Monitoring	
	16	29/09/2017			Retest 6 months: Dielectric and Water Content	

Summary by Soetjahjono
TxM Indonesia